

Bank Term Deposit Subscription Prediction

ADA 442 Project

Yankı Ufuk Dülger

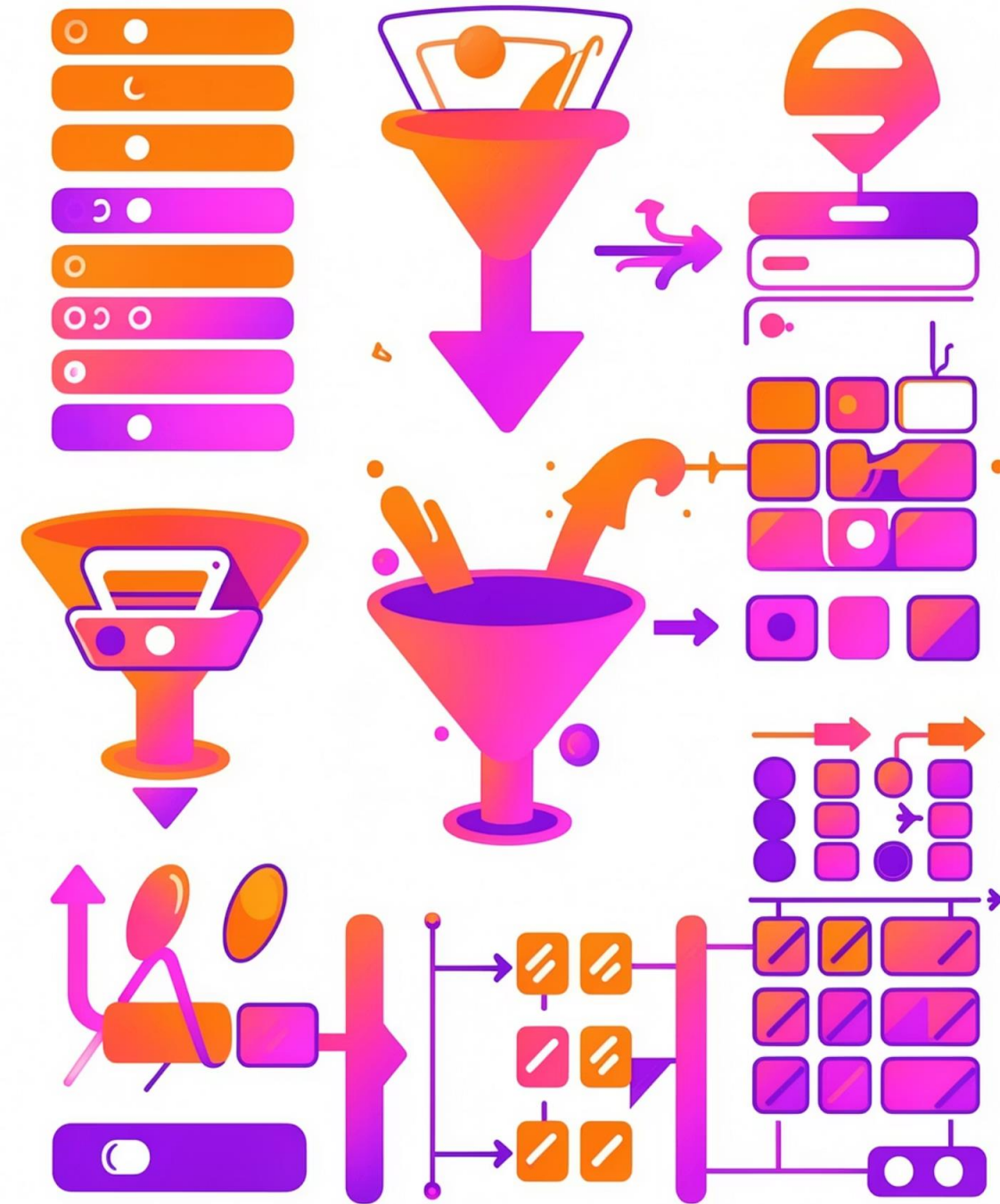
Bank Term Deposit Subscription Prediction

This project aims to predict whether a bank client will subscribe to a term deposit using the Bank Marketing dataset and machine learning classification models.



Data Cleaning and Preprocessing

The dataset was checked for missing values, duplicates, class imbalance, and data leakage. The duration column was removed because it would not be available before a real prediction.



Feature Engineering and Pipeline

New features were created from education, job type, loan status, previous contact, campaign activity, and economic indicators.

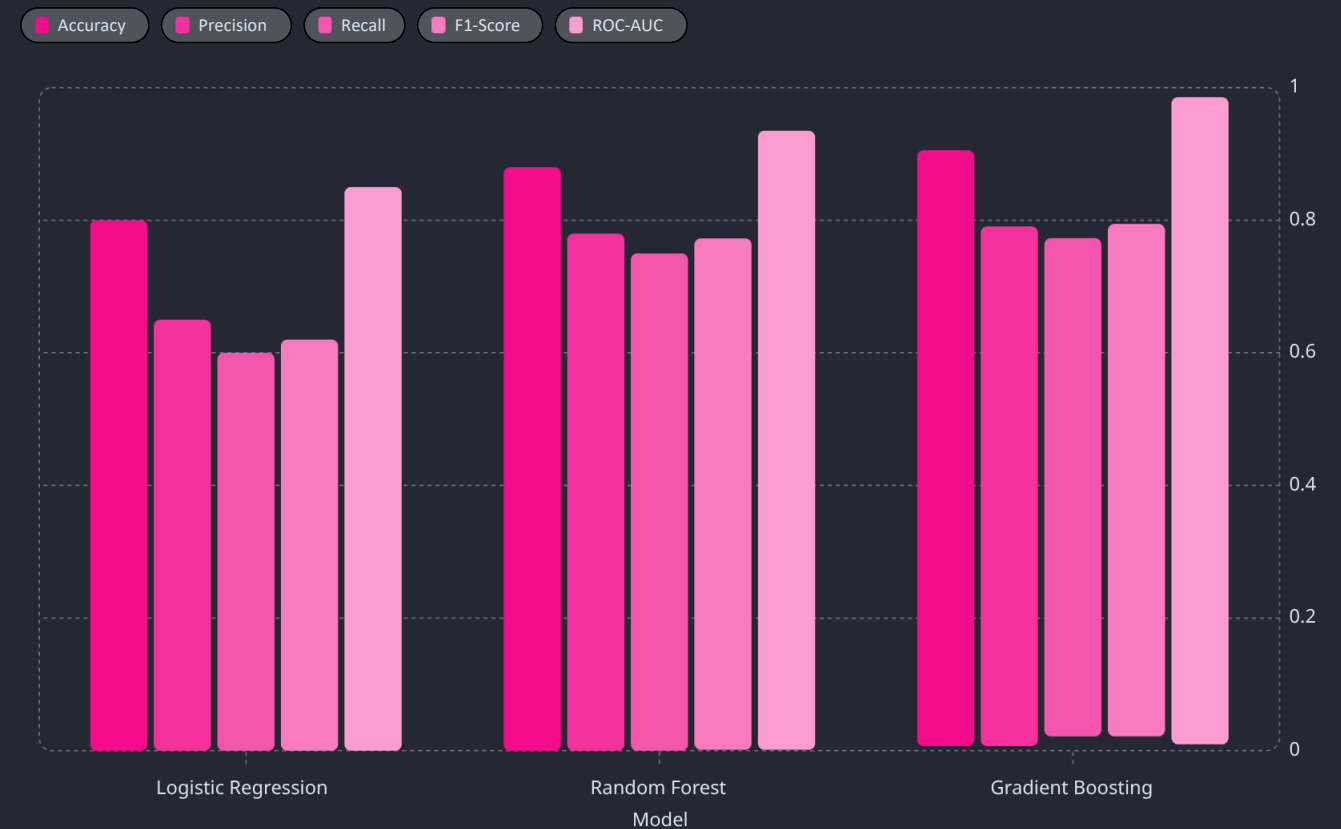
A pipeline was built for preprocessing, feature selection, SMOTE, and model training.



This diagram illustrates the structured flow of our machine learning process, transforming raw data into a trained model through distinct, sequential stages.

Model Training and Evaluation

Logistic Regression, Random Forest, and Gradient Boosting models were compared using accuracy, precision, recall, F1-score, and ROC-AUC. Random Forest was selected and tuned with GridSearchCV.



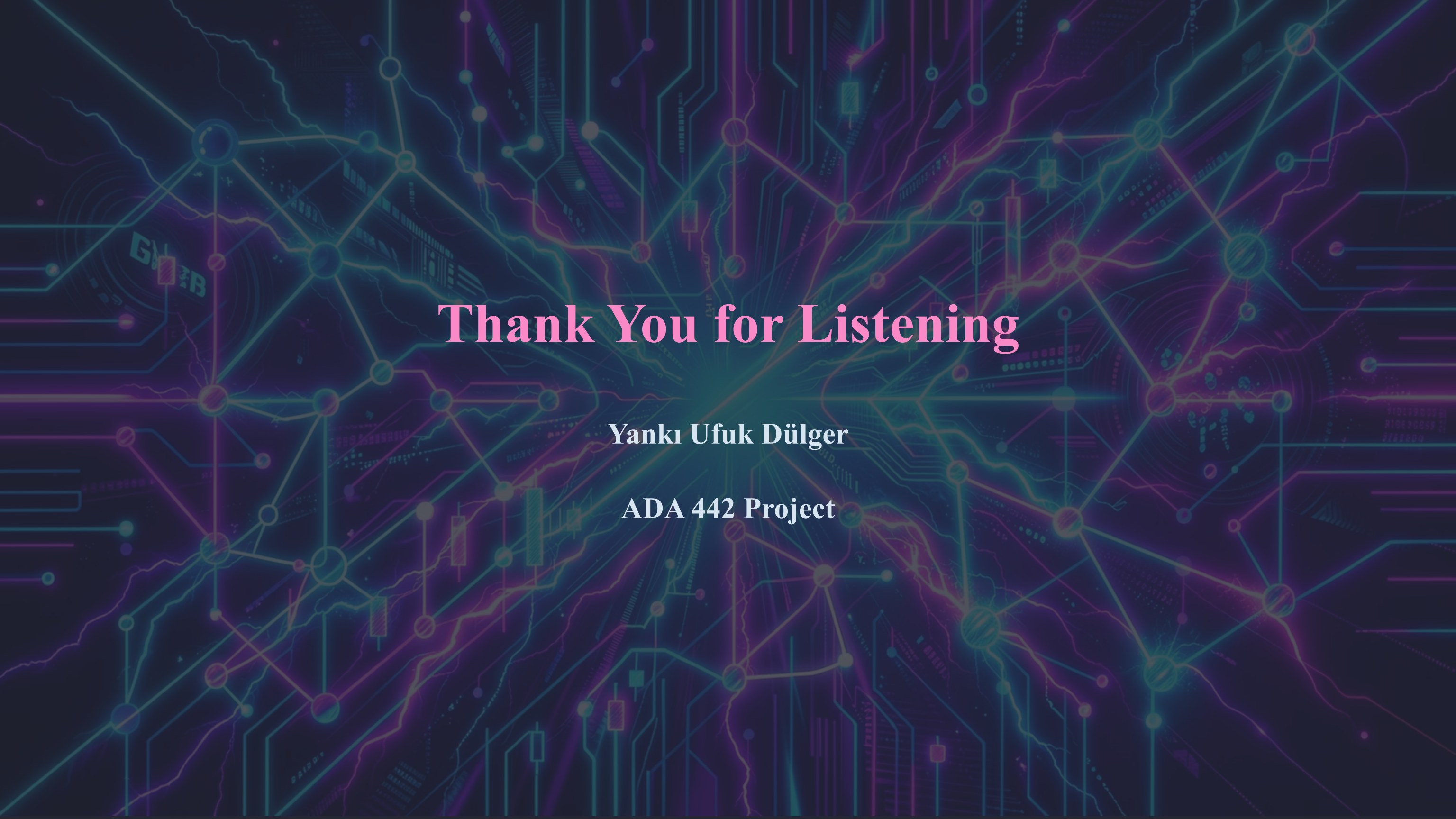
Streamlit Deployment and Conclusion

The final model was deployed with Streamlit Cloud as an interactive web application where users can explore the data and make customer subscription predictions.

Deployment Link:

<https://i4mddnuxhdgtcwutbpmdjm.streamlit.app/>





Thank You for Listening

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